

The De-margining Artifact

*does a forecasting model's measured edge over the bookmaker
survive the choice of margin-removal transform?*

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Abstract

Empirical football-forecasting studies routinely claim an edge over the bookmaker. Substantiating such a claim requires converting offered odds into a benchmark probability, which means removing the bookmaker’s margin. At least four transforms are in use for this: proportional normalisation, the power transform, Shin’s insider-trading model, and the constant odds-ratio transform. Papers typically apply one without justification. The four are not equivalent: they differ chiefly in how much margin they attribute to longshots, which is precisely where a model’s claimed edge tends to concentrate.

We hold a single walk-forward Dixon–Coles model fixed across all four transforms and vary nothing but the benchmark. The comparison covers 176,629 archived matches, 147,177 out-of-sample forecasts, and 91 (league, bookmaker, market) cells drawn from 15 leagues and 14 bookmakers, on 1X2 and Over/Under 2.5 closing prices. Three predictions were pre-registered before the evaluation period was touched: that transform disagreement scales with the book’s margin (P1); that it is larger in the three-outcome market than the two-outcome one (P2); and that there exist cells in which the measured edge changes *sign* between the proportional and Shin benchmarks (P3). P1 is directionally supported: across the 91 cells, Spearman’s rank correlation between a cell’s book margin and the spread across transforms is 0.594. P2 is directionally consistent with the prediction, although it rests on only 15 paired cells. P3 does not hold, in any meaningful sense. Only 1 cell of 91 reverses sign, on returns that are statistically indistinguishable from zero on both sides. The fixed model achieves no edge anywhere in the sample, and loses to the closing line in effectively every cell. The choice of transform demonstrably changes what is measured. Whether it can overturn a published conclusion remains open, and this design cannot settle it.

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1 Introduction

A bookmaker’s quoted prices do not sum to a probability. For a market with outcomes $i = 1, \dots, n$ and offered decimal odds d_i , the raw implied masses $p_i = 1/d_i$ sum to $B = \sum_i p_i > 1$; the excess $B - 1$ is the overround, the bookmaker’s margin. Any study that claims a forecasting edge must first turn $\{p_i\}$ into a probability vector $\{\pi_i\}$ summing to one, because it is against that vector, not against the quoted prices, that the model’s superiority is asserted.

That conversion is a modelling choice, and the literature has not converged on one. Proportional normalisation is the default. Shin’s model derives a transform from the presence of insider money [25, 26]. Štrumbelj [27] compared those two as probability estimators against two logit regressions, and found Shin more accurate than basic normalisation for every bookmaker/sport pair (Section 2). The power transform [5] and the constant odds-ratio transform [4] were not in that comparison. The four coincide only in the limit $B \rightarrow 1$. Away from it they differ in how the removed mass is distributed across outcomes, and they differ most at long odds. This matters because the favourite–longshot bias concentrates market mispricing at exactly those prices [3], and because a model’s largest apparent disagreements with the market are usually found there too.

The question this paper asks is therefore not whether a model can beat the market. It is narrower and, we argue, prior to it:

Does the measured edge of a fixed model over the bookmaker depend materially on which de-margining transform constructs the benchmark?

If the answer is no, the field’s casual treatment of the choice is harmless. If the answer is yes, and in particular if the sign of a measured edge can be flipped by substituting one standard transform for another, then a class of published results is under-determined by its own method section. Reporting the transform then becomes a minimum requirement rather than a courtesy.

Our contribution is threefold. First, we isolate the transform as the sole experimental factor: one model, one fit per matchday, one betting rule, four benchmarks. Second, we pre-register the predictions and the frozen model hyper-parameter before the evaluation period is run, so the git history of the pre-registration timestamps the hypotheses ahead of the results (Section 6). Third, we report the full coverage matrix, including every excluded cell, so the reader can see what the study could not measure as well as what it could.

2 Related work

The question this paper isolates sits at the intersection of three literatures that rarely cite one another: how a bookmaker’s margin should be removed, how a football match should be forecast, and how the resulting forecasts should be compared against the market. Each is summarised in turn, and each summary ends by saying what it does, and does not, settle for the design used here.

2.1 Removing the margin, and where it goes

Overround exists because the bookmaker prices risk, not merely outcomes, and the earliest formal account of why is Shin’s: modelling a market maker who faces bettors who may be privately informed, Shin derives the odds a risk-neutral bookmaker should set to protect against insider money and, by inverting the relationship, a rule for recovering fair probabilities from observed odds [25, 26]. The rule requires solving for a single latent parameter, z , interpreted as the fraction of turnover originating from insiders. The same z later gets used as a measure of market’s informational efficiency in its own right, not merely as a by-product of de-margining. Cortis derives the corresponding accounting identity from the bookmaker’s side, giving closed-form

expressions for the mean and variance of payout as a function of margin and stake distribution, which is the complementary result to Shin’s demand-side model: together they describe the same over-round from the two sides of the book [7].

Proportional normalisation has no comparable structural derivation. It makes no assumption about how the margin is allocated across outcomes and preserves the ratio of any two implied probabilities, which is why it is the field’s default. Which of these maps is the most accurate probability forecast of the realised outcome has been studied at scale more than once. Štrumbelj [27] treats the choice explicitly as a design decision and compares basic (proportional) normalisation, Shin, and two logit regressions on 37 competitions across five sports, finding Shin more accurate than normalisation for every bookmaker/sport pair. Goto et al. [15] evaluate multiplicative (proportional), Shin and power conversions on 90,014 football matches across five bookmakers, and propose two further maps: an odds-only equal-profitability method and a favourite–longshot generalised linear model. Their question is which conversion performs best as a forecast. Ours is whether, holding a forecasting model fixed, the scientific conclusion about that model’s edge over the bookmaker depends on the conversion. The two contributions are complementary: a ranking of converters does not settle whether swapping converters can change a model’s measured edge, and a sensitivity study does not settle which converter is the better probability forecast. Neither paper asks whether the choice can reverse a model’s measured edge, which is the gap addressed here.

The reason any of this matters for a forecasting study is the favourite–longshot bias: bettors accept a lower expected return on longshots than on favourites, a regularity documented across parimutuel and fixed-odds markets alike and surveyed at length by Ottaviani and Sørensen [22], who catalogue explanations ranging from risk-loving preferences to informational asymmetry of exactly the kind Shin’s model formalises. Whatever its cause, the bias means the market’s mispricing, and so a model’s largest apparent disagreements with the market, concentrates at long odds [3]. The four transforms compared here differ most exactly there (Section 3), which is the structural reason a transform choice can matter for a claimed edge and not an incidental one.

2.2 Forecasting the match

The Poisson framework used here begins with Maher [19], who modelled each team’s goals scored and conceded as independent Poisson variables governed by attack and defence parameters, recovered by maximum likelihood from a season of scores. Dixon and Coles [9] correct that model’s chief empirical failure, an under-prediction of low-scoring draws, with the τ adjustment used unchanged in this study (Section 5), and add exponential time-weighting so that a team’s estimated strength reflects its recent form rather than an unweighted season average. Both extensions are structural to the model fit here. Neither addresses how the resulting forecast should be benchmarked, which is outside the scope of the original paper and the reason this study exists as a separate exercise.

Two further lines extend Dixon–Coles in directions this study does not take. Rue and Salvesen [24] replace the static per-season attack and defence parameters with a state-space model updated match-by-match via Markov chain Monte Carlo, letting strength drift continuously rather than being refit from a fixed window; Karlis and Ntzoufras [16] replace the independent-Poisson-plus-correction device with a bivariate Poisson distribution that models the scoreline dependence directly through a shared latent component, dispensing with the four-cell patch that τ represents. Both are more sophisticated models of the scoreline than the one used here, and both would very plausibly forecast better. Adopting either would improve the *level* of the model’s accuracy without changing the paper’s design, which needs the forecasting model held fixed rather than optimised: see Section 5 for why, and Section 9 for what a stronger model would and would not add to this study’s conclusion.

2.3 Is the market beatable, and how would you know

Levitt [18] shows that bookmakers behave less like risk-neutral market-clearing intermediaries than like informed traders who set prices to exploit predictable bettor bias while carrying inventory risk themselves. That is one structural reason the closing line is a demanding benchmark, not merely a busy one. Directly testing whether it can be beaten by a statistical model, Forrest et al. [12] compare bookmaker-implied forecasts against an independent statistical model for English football and find the bookmakers' forecasts equal or superior on most measures, a result the present study's Table 6 reproduces in direction, though not because a similarly specified model must always lose. A market can be soft in a segment a study happens not to cover. Štrumbelj and Robnik-Šikonja [28] extend the comparison across ten online bookmakers and six leagues and find material differences in forecast quality *between* bookmakers, which is part of the motivation for reporting results per bookmaker here rather than pooling books together (Table 3). Franck et al. [13] give one reason those differences persist: bookmakers price in sentimental demand as well as expected outcome, shading odds toward popular teams, which is a source of cross-sectional heterogeneity in the margin that this study's design absorbs into the per-cell margin rather than modelling directly.

None of these papers asks the question this one does. They study whether the market can be beaten, or why bookmakers price the way they do. This paper holds the answer to the first question fixed (our model does not beat the market anywhere in the sample: Section 7) and asks whether the *measurement* of an edge, beaten or not, is itself stable under a choice every one of these papers had to make and rarely reports.

2.4 Scoring the forecast

Brier [2] and Epstein [10] give football forecasting its two standard proper scores, and Gneiting and Raftery [14] give the general theory of why propriety (the forecaster is not rewarded for reporting anything other than a genuine belief) is the property worth insisting on. Whether the football literature should prefer Epstein's distance-sensitive ranked probability score over the ordering-blind Brier score is contested rather than settled: Constantinou and Fenton [6] argue for RPS on the grounds that a near-miss forecast (predicting a draw when the away side won by one) should be scored better than a distant one, while Wheatcroft [29] argues that distance-sensitivity rewards nothing a forecast user actually needs and recommends against it on those grounds. This study reports RPS as its primary score, following the majority convention in the sport, but decomposes Brier in Section 5.4 precisely because the Murphy decomposition [20] used there is defined for Brier and has no RPS analogue; nothing in the P1–P3 comparison depends on resolving the RPS-versus-Brier question either way, since both are computed identically across all four transforms and the comparison is paired throughout.

One measure conspicuously absent from the results is a Kelly-optimal stake [17]. Kelly staking sizes each bet by the model's own edge and so entangles the measured return with the size of the (here, generally absent) edge in a way flat staking does not. Section 5.4's one-unit rule is chosen so that the ROI comparison between transforms isolates the benchmark, not the staking rule, at the cost of not reporting the return a Kelly-sized bettor would actually have achieved.

3 The instrument: four de-margining transforms

Each transform maps the raw implied masses $p \in (0,1)^n$, with book sum $B = \sum_i p_i$, to a probability vector π with $\sum_i \pi_i = 1$.

Proportional. Remove margin uniformly in proportion to implied mass:

$$\pi_i = \frac{p_i}{B}. \tag{1}$$

It fits no free parameter and preserves the ratio of any two implied masses.

Power. Solve for the exponent k satisfying

$$\sum_i p_i^k = 1, \quad \pi_i = p_i^k. \quad (2)$$

Because every $p_i \in (0, 1)$, the sum is strictly decreasing in k , equal to n at $k = 0$ and to B at $k = 1$; a margined book therefore has its unique root at $k > 1$. The transform takes proportionally more margin from longshots, in the direction implied by the favourite-longshot bias [5].

Shin. Model a proportion z of the money wagered as coming from insiders, and recover [26]

$$\pi_i = \frac{\sqrt{z^2 + 4(1-z)p_i^2/B} - z}{2(1-z)}, \quad (3)$$

with z chosen so that $\sum_i \pi_i = 1$. Note that as $z \rightarrow 0$ this tends to $p_i/\sqrt{B}$, not to p_i/B : Shin does not nest proportional normalisation at its boundary.

Odds-ratio. Impose a constant ratio c between the offered and true odds of every outcome, $p_i/(1-p_i) = c\pi_i/(1-\pi_i)$, which rearranges to

$$\pi_i = \frac{p_i}{c(1-p_i) + p_i}, \quad (4)$$

with c solving $\sum_i \pi_i = 1$. At $c = 1$ the map is the identity and π_i is decreasing in c , so a margined book has its root at $c > 1$ [4].

3.1 Existence and uniqueness of the fitted parameter

Each of the three fitted transforms is presented above as “solve for k (or z , or c) such that the recovered vector sums to one,” which presupposes that a solution exists and that the solver converges to the right one rather than an arbitrary one. That is worth establishing rather than assuming, since a solver silently converging to a spurious root would corrupt every downstream number without producing an error anywhere.

Proposition 1 (Power). *For $p \in (0, 1)^n$ with $B = \sum_i p_i > 1$, there is exactly one $k^* > 1$ with $\sum_i p_i^{k^*} = 1$.*

Proof. Let $f(k) = \sum_i p_i^k$. Each term $p_i^k = \exp(k \log p_i)$ has derivative $p_i^k \log p_i$, strictly negative since $p_i \in (0, 1)$ implies $\log p_i < 0$; hence f is strictly decreasing and continuous on $\mathbb{R}$. $f(1) = B > 1$ and $f(k) \rightarrow 0$ as $k \rightarrow \infty$ (each term is bounded above by $\max_i p_i^k \rightarrow 0$), so by the intermediate value theorem f crosses 1 exactly once, at some $k^* > 1$ since $f(1) > 1$ and f is decreasing. $\square$

Proposition 2 (Odds-ratio). *For $p \in (0, 1)^n$ with $B > 1$, there is exactly one $c^* > 1$ with $\sum_i p_i / (c^*(1 - p_i) + p_i) = 1$.*

Proof. Write $g(c) = \sum_i p_i / (c(1 - p_i) + p_i)$. Each term has derivative $-p_i(1 - p_i) / (c(1 - p_i) + p_i)^2 < 0$ for $p_i \in (0, 1)$ and $c > 0$, so g is strictly decreasing on $(0, \infty)$. $g(1) = B > 1$, and $g(c) \rightarrow 0$ as $c \rightarrow \infty$ termwise, so as with the power transform a unique root $c^* > 1$ exists by the intermediate value theorem. $\square$

Proposition 3 (Shin). *For $p \in (0, 1)^n$ with $B > 1$, there exists $z^* \in (0, 1)$ solving $\sum_i \pi_i(z^*) = 1$, where $\pi_i(z)$ is given by Equation (3).*

Proof. Write $h(z) = \sum_i \pi_i(z)$. At $z \rightarrow 0^+$, $\pi_i(z) \rightarrow p_i / \sqrt{B}$ termwise (Section 3), so $h(0^+) = B / \sqrt{B} = \sqrt{B} > 1$. As $z \rightarrow 1^-$, write $s = 1 - z$ and expand $\sqrt{z^2 + 4s p_i^2 / B} = \sqrt{(1 - s)^2 + 4s p_i^2 / B}$ to first order in s : the term is $1 - s + 2s p_i^2 / B + O(s^2)$, so $\pi_i(z) = [(1 - s + 2s p_i^2 / B) - (1 - s)] / (2s) + O(s) \rightarrow p_i^2 / B$. Hence $h(1^-) = \sum_i p_i^2 / B$. Since $0 < p_i < 1$ for every i , $p_i^2 < p_i$ termwise, so $\sum_i p_i^2 < \sum_i p_i = B$ and $h(1^-) < 1$. h is continuous on $(0, 1)$, so by the intermediate value theorem it crosses 1 somewhere in $(0, 1)$, giving existence of z^* . Monotonicity of h , which gives uniqueness rather than merely existence, is established in Shin [26]. The $z \rightarrow 0^+$ boundary is exercised directly by the test suite (`test_shin_limit_as_z_vanishes_is_p_over_sqrt_b` in `tests/test_demargin.py`), which exists because an earlier version of this code used the wrong limit, p_i / B , a bug that summed to exactly one and so produced no error anywhere downstream. The $z \rightarrow 1^-$ boundary above was checked against the implementation directly for this paper rather than asserted in the suite. $\square$

The bisection driver used for all three (Section 3.3) does not rely on these propositions to run. It will return *some* value whether or not a root exists in the bracket it is given, so the brackets themselves are chosen wide enough that existence is never in doubt ($[1, \infty)$ enlarged geometrically for k and c ; $[0, 1 - 10^{-12}]$, which brackets a sign change by the argument above, for z), and the propositions are what justify treating the bisection's output as *the* answer rather than *an* answer.

3.2 A closed form for two-outcome markets

The Over/Under 2.5 market used throughout this study (Section 4) has $n = 2$, and at $n = 2$ Shin's transform simplifies enough to solve without a root-find at all. Clarke et al. [5] already record that Shin and additive margin removal coincide for two-outcome markets. We record the closed form and show why the fitted insider fraction z drops out of the recovered probabilities, which also explains part of why P2 (Section 7) finds smaller transform disagreement in the two-outcome market than the three-outcome one.

Proposition 4. *For $n = 2$, write $p = (p_1, p_2)$, $B = p_1 + p_2 > 1$, $d = p_1 - p_2$. The Shin probabilities do not depend on the fitted parameter z and equal*

$$\pi_1 = \frac{1 + d}{2} = p_1 - \frac{B - 1}{2}, \quad \pi_2 = \frac{1 - d}{2} = p_2 - \frac{B - 1}{2}. \quad (5)$$

Proof. Summing the Shin equation over $i = 1, 2$ and clearing the common denominator $2(1 - z)$ gives $\sqrt{z^2 + 4(1 - z)p_1^2 / B} + \sqrt{z^2 + 4(1 - z)p_2^2 / B} = 2$: the two linear-in- z terms that appear for general n collapse to a constant precisely because $n = 2$. Write $s = 1 - z$, $u = \sqrt{z^2 + 4s p_1^2 / B}$, $v = \sqrt{z^2 + 4s p_2^2 / B}$, so $u + v = 2$. Since $B = p_1 + p_2$,

$$u^2 - v^2 = \frac{4s(p_1^2 - p_2^2)}{B} = \frac{4s(p_1 - p_2)(p_1 + p_2)}{B} = 4sd,$$

and $u - v = (u^2 - v^2)/(u + v) = 2sd$. Solving the pair $u + v = 2$, $u - v = 2sd$ gives $u = 1 + sd$ and $v = 1 - sd$. Substituting into $\pi_1 = (u - z)/(2s)$ with $z = 1 - s$:

$$\pi_1 = \frac{(1 + sd) - (1 - s)}{2s} = \frac{s(1 + d)}{2s} = \frac{1 + d}{2},$$

and symmetrically $\pi_2 = (1 - d)/2$. Neither expression involves s or z : whichever $z^* \in (0, 1)$ solves the summed equation (Proposition 3), it produces the same pair of probabilities, so the value of z^* itself is irrelevant to the recovered vector at $n = 2$. Rewriting $(1 + d)/2$ as $p_1 - (p_1 + p_2 - 1)/2 = p_1 - (B - 1)/2$ gives the stated closed form. Verified against the bisection output on 6 spot-checked books to a relative error below 10^{-12} . $\square$

At $n = 2$, then, Shin is exactly *additive* margin removal: split the excess $B - 1$ evenly and subtract it from each raw implied probability, independently of the insider-trading story that motivates the general- n model. Proportional, power and odds-ratio remain *multiplicative* at every n , including $n = 2$ (Table 4 shows they still disagree with each other and with Shin there). The transform that models the favourite-longshot asymmetry most explicitly is, at exactly two outcomes, the one transform that cannot express any asymmetry between the two outcomes' margin at all: $\pi_1 - \pi_2 = p_1 - p_2$ is preserved exactly, whatever the margin. That is a structural reason, not merely an empirical one, for P2's finding that disagreement is smaller in the two-outcome market.

3.3 Numerical solution: a common bisection driver

Three of the four require a root-find. To ensure that no transform is advantaged by a looser solver, all four share a single bisection driver at a common tolerance of 10^{-12} with a common iteration budget, and each transform's recovered vector is renormalised identically afterwards. The fitted free parameter ($k, z, c; B$ for proportional, which fits nothing) is retained per market so that the effort each transform expended on a given book can be reported. Appendix B walks through the driver's implementation and the numerical safeguards around it.

4 Data

Match results and bookmaker odds come from Football-Data.co.uk [11], which publishes per-league, per-season CSV files containing full-time scores and the closing prices of a panel of bookmakers, and, for recent seasons, the opening prices as well.

525 season files were fetched and 514 parsed, yielding 176,629 matches. The eleven unreadable files are recorded rather than silently dropped: seven declare an internal division code that disagrees with their filename, three carry structurally corrupted rows, and one declares the same bookmaker column twice. An unreadable league-season is missing data, and missing data nobody counted is how a coverage table becomes a lie.

Two markets are used: the three-outcome match-odds market (1X2) and the two-outcome Over/Under 2.5 goals market. *Closing* prices only. Opening prices are never used as a benchmark, because a benchmark drawn before the market has absorbed team news measures something other than the market's final opinion. A match survives into a cell only if its closing book is *complete* (every outcome priced), since a transform applied to a partial book normalises over the wrong support and returns a benchmark that is wrong rather than merely noisy.

The unit of analysis is the (league, bookmaker, market) cell. Cells with fewer than 200 matched fixtures are excluded. The threshold was fixed in the pre-registration rather than chosen after inspection. The full coverage matrix is published as a result, including every excluded cell (Table 1).

Table 1: Coverage. Complete closing books per bookmaker and market, and the cells that clear the 200-fixture threshold.

Bookmaker	Market	Cells	Matches
Pinnacle	1X2	15	30,769
Bet365	1X2	5	9,838
Bet365	OU25	5	9,838
Bet&Win	1X2	5	9,269
Pinnacle	OU25	5	8,916
William Hill	1X2	5	7,718
VC Bet	1X2	5	7,002
Interwetten	1X2	5	6,243
Betfair Exchange	1X2	5	2,729
Betfair Exchange	OU25	5	2,728
Betfair	1X2	5	1,412
1XBet	1X2	5	1,412
Betfred	1X2	5	1,383
BetVictor	1X2	5	1,371
BetMGM	1X2	5	1,341
Ladbrokes	1X2	3	677
Coral	1X2	3	643

5 Method

5.1 Forecasting model

The model is Dixon–Coles [9], building on the independent-Poisson scoring model of Maher [19]: independent Poisson scoring rates with a low-score dependence correction. For a match between home team h and away team a ,

$$\lambda = \alpha_h \beta_a \gamma, \quad \mu = \alpha_a \beta_h, \quad (6)$$

where α is attack strength, β defence weakness, and γ a per-league home advantage. The joint mass of a scoreline (x, y) is

$$P(x, y) = \tau(x, y, \lambda, \mu, \rho) \frac{e^{-\lambda} \lambda^x}{x!} \frac{e^{-\mu} \mu^y}{y!}, \quad (7)$$

with τ the Dixon–Coles low-score correction, a piecewise adjustment applied to the four scorelines in which both sides score at most one and left at 1 elsewhere:

$$\tau(x, y, \lambda, \mu, \rho) = \begin{cases} 1 - \lambda\mu\rho & (x, y) = (0, 0) \\ 1 + \lambda\rho & (x, y) = (0, 1) \\ 1 + \mu\rho & (x, y) = (1, 0) \\ 1 - \rho & (x, y) = (1, 1) \\ 1 & \text{otherwise.} \end{cases} \quad (8)$$

A single dependence parameter ρ governs the correction, and $\rho < 0$ inflates $\tau(0, 0)$ and $\tau(1, 1)$, correcting independent Poisson’s chief empirical failure, an under-prediction of low-scoring draws. Not every ρ is admissible at a given (λ, μ) : τ must stay non-negative at all four corrected cells, which requires

$$\rho \in \left[\max \left(-\frac{1}{\lambda}, -\frac{1}{\mu} \right), \min \left(\frac{1}{\lambda\mu}, 1 \right) \right]. \quad (9)$$

This region depends on the fixture’s own rates, which is why a ρ valid for every match in a fit window can still fall outside it for a pairing not in that window (Section 5.2).

Fitting a matchday’s model is weighted maximum likelihood over the log-scale parameter vector $\theta = (\log \alpha, \log \beta, \log \gamma, \rho)$, with match m weighted by $w_m = \exp(-\xi t_m)$ (Section 5.3) for its age t_m in days at the fit cutoff. Dropping the outcome-independent factorial terms, the objective is

$$\ell(\theta) = \sum_m w_m \left[\log \tau(x_m, y_m, \lambda_m, \mu_m, \rho) + x_m \log \lambda_m - \lambda_m + y_m \log \mu_m - \mu_m \right], \quad (10)$$

with λ_m, μ_m given by scaling each team’s fitted attack and defence parameters as above. The log parameterisation makes positivity structural rather than a bound and gives $\partial \lambda_m / \partial \log \alpha_{h(m)} = \lambda_m$ directly, which is what makes the gradient below closed-form rather than merely differentiable in principle. Two further choices matter for identifiability and are stated rather than left implicit: (i) scaling every α by a constant c and every β by $1/c$ leaves every λ_m, μ_m unchanged, so the likelihood has a flat direction. It is removed by dropping one team’s log-attack parameter from the free vector and setting it to minus the sum of the rest, fixing $\sum_i \log \alpha_i = 0$; (ii) the optimiser is started at $\rho_0 = -0.10$ rather than 0, because the $\tau(0, 1)$ and $\tau(1, 0)$ derivatives with respect to ρ vanish identically at $\rho = 0$ (Equation (11) below), so starting exactly there gives the optimiser no first-order signal about ρ from those two cells.

Maximising Equation (10) needs its gradient, and needs it several thousand times over the course of one walk-forward run (Section 5.2): one refit per matchday per league, roughly 40 matchdays a season across 33 seasons and 16 leagues. A finite-difference gradient costs one likelihood evaluation per parameter, on the order of 80 for a 20-team league, once attack, defence, home advantage and ρ are counted, so an analytic gradient is not a micro-optimisation here but the difference between a run measured in minutes and one measured in hours. Differentiating Equation (8) at the four corrected cells and zero elsewhere gives

$$\frac{\partial \tau}{\partial \lambda} = \begin{cases} -\mu\rho & (0, 0) \\ \rho & (0, 1) \\ 0 & \text{else,} \end{cases} \quad \frac{\partial \tau}{\partial \mu} = \begin{cases} -\lambda\rho & (0, 0) \\ \rho & (1, 0) \\ 0 & \text{else,} \end{cases} \quad \frac{\partial \tau}{\partial \rho} = \begin{cases} -\lambda\mu & (0, 0) \\ \lambda & (0, 1) \\ \mu & (1, 0) \\ -1 & (1, 1) \\ 0 & \text{else.} \end{cases} \quad (11)$$

Writing the per-match contribution to $\partial \ell / \partial \log \lambda_m$ and $\partial \ell / \partial \log \mu_m$ using $\partial \lambda_m / \partial \log \lambda_m = \lambda_m$,

$$s_m^\lambda = w_m \left[\frac{\lambda_m}{\tau_m} \frac{\partial \tau_m}{\partial \lambda_m} + x_m - \lambda_m \right], \quad s_m^\mu = w_m \left[\frac{\mu_m}{\tau_m} \frac{\partial \tau_m}{\partial \mu_m} + y_m - \mu_m \right], \quad (12)$$

the gradient with respect to each team’s log-attack and log-defence parameter is a sum of s_m^λ and s_m^μ over exactly the matches that team played at home or away,

$$\frac{\partial \ell}{\partial \log \alpha_i} = \sum_{m: h(m)=i} s_m^\lambda + \sum_{m: a(m)=i} s_m^\mu, \quad \frac{\partial \ell}{\partial \log \beta_i} = \sum_{m: a(m)=i} s_m^\lambda + \sum_{m: h(m)=i} s_m^\mu, \quad (13)$$

with the two remaining parameters collecting a sum over every match rather than a per-team subset,

$$\frac{\partial \ell}{\partial \log \gamma} = \sum_m s_m^\lambda, \quad \frac{\partial \ell}{\partial \rho} = \sum_m w_m \frac{\partial \tau_m / \partial \rho}{\tau_m}. \quad (14)$$

Every sum in Equations (13) and (14) is a scatter-add keyed by team index, computed without a Python-level loop over matches (`numpy.bincount`; Appendix B). A naive implementation looping over teams and matches is quadratic in the two. The scatter-add form is a single linear pass over the matches instead. The gradient with respect to the dropped attack parameter is folded into the free entries by the chain rule implied by constraint (i) above: $\partial \ell / \partial (\text{free } \log \alpha_i) =$

$\partial\ell/\partial\log\alpha_i - \partial\ell/\partial\log\alpha_{\text{last}}$. The whole gradient is verified against `scipy.optimize.check_grad` in the test suite on every commit, because a subtly wrong analytic gradient converges silently to the wrong optimum: nothing downstream (not the likelihood value, not the fitted probabilities, not any later test) would flag that the optimum reached was wrong rather than merely different.

The model choice is deliberate and, for this study, incidental. A better model (Rue and Salvesen’s dynamically updated ratings, or Karlis and Ntzoufras’s bivariate Poisson, Section 2) would very plausibly be better, and would change the level of every measured edge. The question here is whether the *measurement* moves when only the benchmark is swapped, and that requires the model to be held fixed, not to be good.

5.2 Walk-forward protocol

Each league’s seasons are partitioned in time into three disjoint periods, fixed before any fitting:

Period	Span	Use
Burn-in	first 3 seasons	fitting only; never forecast or scored
Calibration	next 2 seasons	used <i>solely</i> to select ξ
Evaluation	all remaining	the study

Within the scored periods the protocol is: before each matchday, refit on every match that kicked off *strictly* earlier, weighted by $w_m = \exp(-\xi t_m)$ where t_m is the match’s age in days at the cutoff; predict that matchday, then roll forward. Burn-in matches remain in the fit window throughout, excluded from scoring, not from estimation. A matchday whose fit window holds fewer than 300 matches is skipped.

The strict inequality is the single point at which look-ahead leakage could enter, so it is enforced in one place in the code and asserted directly by the test suite.

A fixture is *dropped rather than priced* in three circumstances: a team absent from the fit window has no fitted strength (a promoted side); a fitted rate falls outside (0.05, 6.0) expected goals, which indicates an unidentified parameter rather than a strong team; or ρ is inadmissible for that fixture’s own (λ, μ) . Admissibility depends on the fixture’s own rates, so a ρ valid for every match in the fit window can still be invalid for a pairing not in it, and the fit-time check is structurally unable to catch that.

5.3 The decay parameter, and how weakly it is identified

ξ was selected by minimising mean ranked probability score over the calibration period only, then frozen at

$$\xi = 0.0017328679513999 \quad (\text{half-life } \ln 2/\xi = 400 \text{ days}). \quad (15)$$

Selection was *paired*. Candidates do not price the same fixtures: a fit can fail to converge, or leave a team unidentified, and an unpaired sweep compared means over match sets differing by 43% in size (7,804 to 11,152 forecasts), which is not a valid comparison: an arm that happens to drop harder fixtures scores better for reasons unrelated to forecast quality. That sweep placed its minimum at 700 days, on the edge of its grid. Scoring every candidate on the 10,567 fixtures that *all* candidates forecast moved the optimum to 400 days and made it interior, with the neighbouring candidates worse on either side (Table 2 and Figure 1).

Stated plainly: ξ is weakly identified. Half-lives of 400 and 700 days differ by 7.3×10^{-5} in paired RPS, and anything between roughly 250 and 1200 days scores within 6×10^{-4} of the optimum. We do not present 400 days as a precisely determined quantity. What Table 2 and Figure 1 establish is that the extremes are wrong. A 30-day half-life is far worse (0.222271), and no decay at all is worse than moderate decay (0.206014 against 0.204889), so exponential decay earns its place without its exact rate mattering much. Because the same model is held fixed across transforms, a weakly identified ξ cannot generate a difference between them.

Table 2: Paired selection of the decay rate on the calibration period. Every candidate is scored on the same 10,567 fixtures.

Half-life (days)	ξ	Paired mean RPS
30	0.023 105	0.222 271
60	0.011 552	0.211 939
90	0.007 702	0.208 691
150	0.004 621	0.206 294
250	0.002 773	0.205 119
400	0.001 733	0.204 889
700	0.000 990	0.205 109
1200	0.000 578	0.205 406
2000	0.000 347	0.205 630
40000	0.000 017	0.206 014

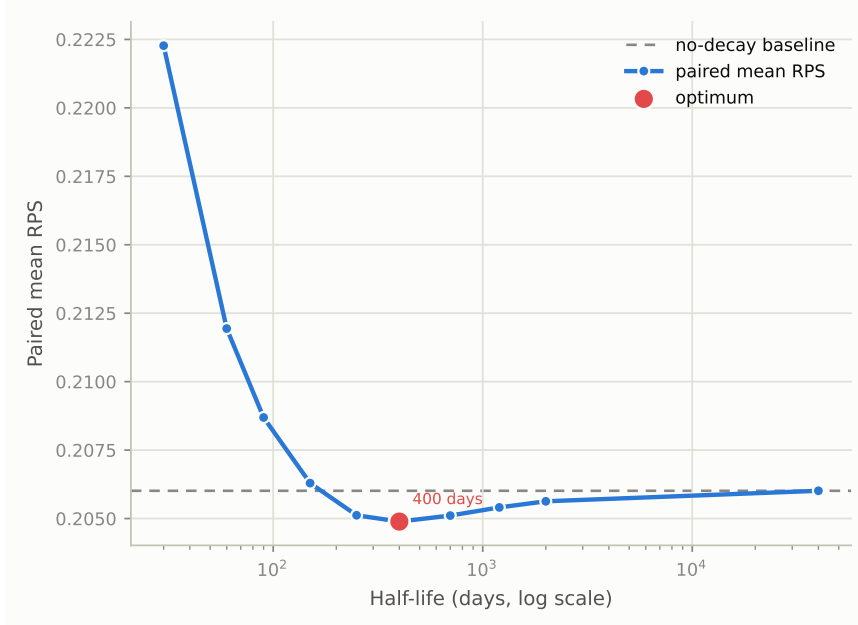


Figure 1: Paired mean RPS against the decay half-life, log scale, for every candidate in the calibration sweep. The marked optimum sits at an interior half-life rather than at either edge of the grid, and the curve stays within a narrow band of the no-decay baseline over a wide neighbourhood around it: ξ is chosen, but only weakly identified, exactly as Table 2 states in numbers.

5.4 Metrics

Ranked probability score. Forecast accuracy is scored with the ranked probability score [10], which respects the ordering of 1X2 outcomes and is the appropriate proper score for this market [6, 14]. For a forecast π over n ordered categories and realised category j ,

$$\text{RPS} = \frac{1}{n-1} \sum_{k=1}^{n-1} \left(\sum_{i \leq k} \pi_i - \mathbf{1}[j \leq k] \right)^2. \quad (16)$$

Both the model and each de-margined benchmark are scored on the same fixtures, so the comparison is paired by construction.

Economic edge. Prediction P3 is an economic statement, so it is measured economically. The betting rule is deliberately minimal: back an outcome when the model’s probability exceeds the de-margined market probability for that outcome, stake one unit, and settle at the *offered* price. Only the benchmark varies between arms. Varying the strategy as well would make any difference in measured edge unattributable. Return on investment is total profit over total turnover, and is reported as undefined, not zero, when a transform selected no bets, since “no bets placed” and “bets placed that broke even” are different findings.

Changing the transform therefore changes the *set of bets*, not merely the measurement of a fixed portfolio: an outcome that clears $p_i^{\text{model}} > p_i^{\text{book},T}$ under proportional normalisation may fail it under Shin, and conversely. That is the economically relevant object — it is how the benchmark affects an actionable edge — but it is not a transform-sensitivity of a held-fixed portfolio. The complementary measure, which scores every fixture and involves no selection, is the ranked probability score already reported alongside ROI (Tables 3–6). P1 and P2 are stated on both.

Score decomposition. Where the source of a difference in accuracy is at issue, the Brier score is decomposed following Murphy [20]. We use the exact five-term form

$$\text{BS} = \text{REL} - \text{RES} + \text{UNC} + \text{WBV} - 2 \text{COV}, \quad (17)$$

whose last two terms (within-bin forecast variance and within-bin forecast–outcome covariance) are identically zero only for single-valued bins. On binned continuous forecasts they are not; see Appendix A.

5.5 Inference

Three tools, each chosen because the naive alternative is invalid on this data rather than merely less powerful.

Forecast comparisons use the Diebold–Mariano test [8] with Newey–West HAC standard errors [21] at the standard truncation lag $\lfloor 4(n/100)^{2/9} \rfloor$. Forecasts produced by overlapping fit windows are serially correlated, and treating per-match score differences as independent would understate the variance and overstate significance.

Interval estimates for ROI use a stationary block bootstrap [23] with geometric blocks of mean length 20 and a fixed seed. Betting returns are heavy-tailed and clustered by matchday. An i.i.d. bootstrap would resample away exactly the dependence that makes the true interval wide.

The study runs a family of tests across leagues, bookmakers, markets and transform pairs, so p -values are controlled for false discovery rate by the Benjamini–Hochberg procedure [1] at $q = 0.10$ across that family.

6 Pre-registration

The predictions, the protocol, the exclusion threshold and the frozen ξ were committed to the repository before the evaluation period was run. The git timestamp on `paper/preregistration.md` is the claim. If that file’s history shows modification after the first evaluation run, every prediction below should be treated as post-hoc. The evaluation script parses ξ out of the committed pre-registration and refuses to start unless it agrees with the calibration output to 10^{-9} . A rounded value in the document would be a *different* decay rate from the one selected, and registering one number while running another would defeat the point of registering it.

ID	Prediction	Falsified by
P1	Disagreement between transforms scales with the book’s margin: negligible on a $\approx 102\%$ book, material on a $\approx 107\%$ book	Transform spread on high-margin cells not exceeding that on low-margin cells
P2	Disagreement is larger for three-outcome 1X2 than for two-outcome Over/Under 2.5	OU25 spread matching or exceeding 1X2 spread on the same cells
P3	There exist (league, bookmaker) cells where the measured edge <i>changes sign</i> between proportional and Shin	No cell showing a sign reversal in ROI between those two transforms

P3 is the headline: sign reversal, not merely magnitude change, is what would invalidate a published conclusion. We committed in advance to reporting all three predictions with effect sizes and confidence intervals regardless of outcome, to dropping or reframing none of them after seeing results, and to writing a null on P3 with P1 and P2 holding as a complete paper.

7 Results

The evaluation produced 147,177 out-of-sample forecasts over 91 cells spanning 15 leagues and 14 bookmakers.

7.1 P1: disagreement scales with the book’s margin

P1 is directionally supported. Across the 91 cells, Spearman’s rank correlation between a cell’s mean book sum and the spread of the four transforms within that cell is 0.594 (ordinary pair-bootstrap 95% interval $[0.454, 0.704]$). That interval treats cells as independent, which they are not: the same bookmakers, and sometimes the same matches, appear in multiple cells. A bookmaker-clustered bootstrap of the same coefficient, which resamples books rather than cells, gives $[0.085, 0.715]$, which still excludes zero. Aggregated to one point per bookmaker — the coarser and more conservative sample — the relationship is 0.442 (95% interval $[-0.154, 0.831]$, $n = 14$), which is wide enough to include zero (Figure 2). The pattern in Table 3 is monotone at the extremes: the two sharpest books, Betfair Exchange at a mean sum of 1.008 and Pinnacle at 1.031, both show a mean RPS spread of 0.0001, while books between 1.06 and 1.09 show 0.0004 to 0.0009. The transforms agree almost exactly where there is little margin to remove and diverge by four to nine times as much where there is more of it, which is what the limiting argument predicts.

The stated falsification condition was that the spread on Bet365 should fail to exceed the spread on Pinnacle. It does exceed it, on both measures (0.0110 against 0.0086 in ROI). The registered contrast is therefore in the predicted direction; the evidence for a precisely estimated correlation is not.

Two cautions. The ROI spread column is noisier than the RPS column and does not order as cleanly, because ROI is a function of how many bets a cell generated and the smallest cells here rest on three league-seasons. And the mechanism is not in doubt (the transforms provably coincide as the book sum approaches one), so what this measures is the size of the effect at realistic margins, not whether it exists.

Figure 2 repeats the comparison at one point per bookmaker, the coarser aggregation behind 0.442. That bookmaker-level ρ need not match the cell-level coefficient above; both are positive and both put the higher-margin books toward the higher-spread end. Figure 3 shows the same disagreement in probability space, on the single highest-margin closing book, chosen by margin alone. The annotated longshot is where the four transforms separate.

Table 3: Transform disagreement against book margin. Spread is the range of a quantity across the four transforms within a cell.

Bookmaker	Cells	Mean book sum	Mean ROI spread	Mean RPS spread
Betfair Exchange	10	1.0080	0.0085	0.0001
Pinnacle	20	1.0309	0.0086	0.0001
1XBet	5	1.0440	0.0317	0.0004
Coral	3	1.0575	0.0489	0.0005
Interwetten	5	1.0624	0.0291	0.0005
Bet365	10	1.0625	0.0110	0.0003
Ladbrokes	3	1.0638	0.0632	0.0004
VC Bet	5	1.0675	0.0293	0.0005
Bet&Win	5	1.0679	0.0236	0.0005
BetMGM	5	1.0744	0.0515	0.0004
BetVictor	5	1.0760	0.0536	0.0004
William Hill	5	1.0761	0.0202	0.0004
Betfred	5	1.0830	0.0520	0.0005
Betfair	5	1.0898	0.0294	0.0009

7.2 P2: disagreement is larger in the three-outcome market

P2 is directionally consistent with the prediction, on thin evidence. On the cells where the same bookmaker prices both markets, the mean ROI spread is 0.0111 for 1X2 against 0.0075 for Over/Under 2.5, and the 1X2 spread is the larger in 60% of pairs on ROI and 80% on RPS.

The evidence is thin and should be read as such. Only three bookmakers in the archive carry closing prices for both markets, which leaves 15 paired cells. The direction matches the prediction and the mechanism is clear: the transforms differ chiefly in how they treat a longshot, and a near-even two-outcome book does not have one. 15 pairs cannot distinguish a moderate effect from a small one. A paired bootstrap of the mean ROI-spread difference (1X2 minus Over/Under) is 0.0036 (95% interval [0.0003, 0.0071]). That interval excludes zero, but it treats the 15 (league, bookmaker) pairs as independent, which they are not: only three bookmakers price both markets. It describes this sample; it is not a licence to generalise.

Table 4: Transform disagreement by market, on cells where both markets are available from the same bookmaker.

Quantity	Pairs	1X2	OU25	Difference	Share 1X2 larger
ROI spread	15	0.0111	0.0075	0.0036	0.6000
RPS spread	15	0.0002	0.0001	0.0001	0.8000

7.3 P3: sign reversal of the measured edge

P3 is met on its letter and fails on its substance, and the failure is the more informative of the two.

Exactly 1 cell of 91 (1.1%) reverses the sign of measured ROI between proportional normalisation and Shin’s method. On the prediction as written (*there exist* cells where the edge changes sign), that is a positive result. Table 5 shows why it should not be read as one.

The cell is B1/Coral/1X2 on 214 matches, barely above the 200-fixture threshold fixed in the pre-registration. Proportional normalisation puts the ROI at +0.0020 and Shin at −0.0004. Both bootstrap intervals span approximately ± 0.10 . These are not two different conclusions about an edge. They are the same conclusion (no measurable edge), with noise falling on opposite sides of zero. A reversal between two quantities that are each indistinguishable from zero carries no information about whether the choice of transform can overturn a finding.

The deeper problem is that the premise of P3 does not hold in this data, for reasons set out in the next section: the model has no edge anywhere, so there is no edge whose sign could be

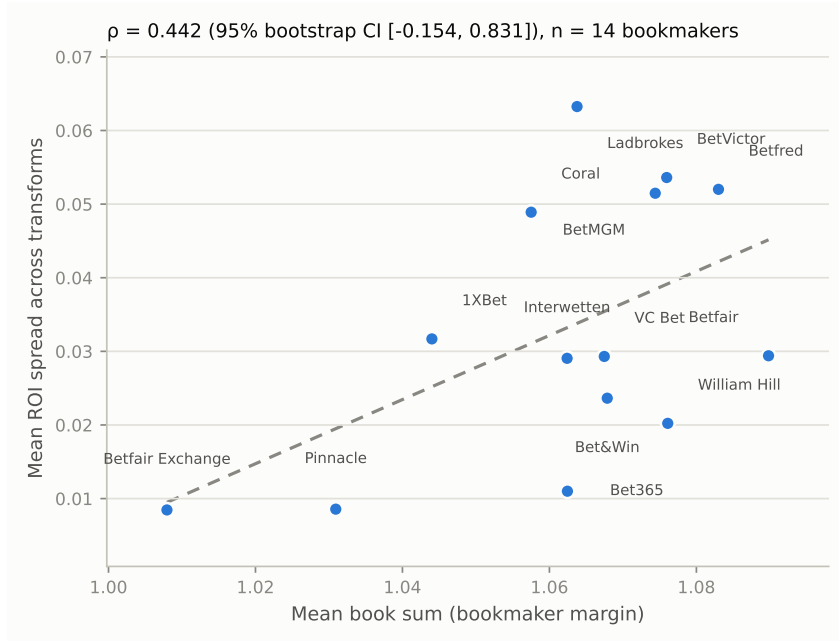


Figure 2: Mean bookmaker margin against mean ROI spread across the four transforms, one point per bookmaker, with a least-squares trend. ρ and its 95% bootstrap interval are computed on these $n = 14$ bookmaker-level points.

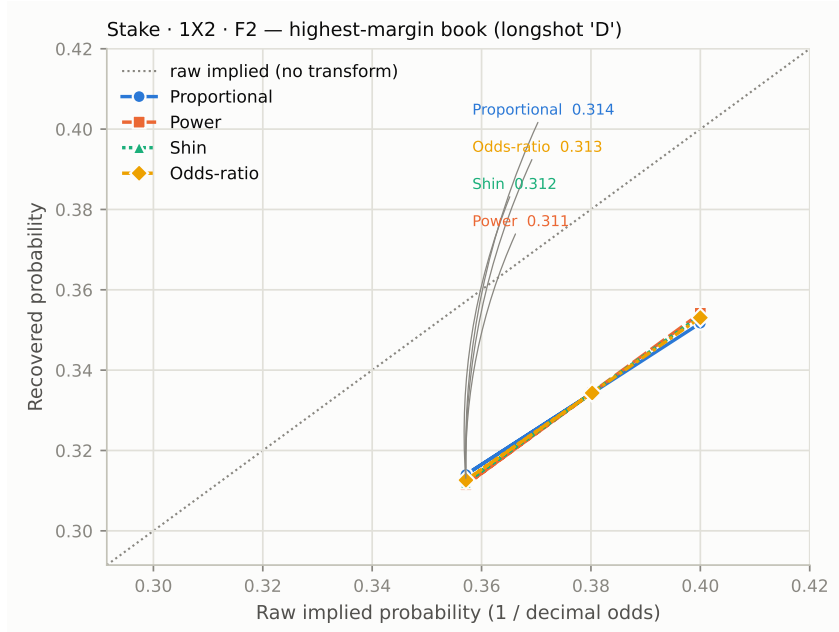


Figure 3: The four transforms' recovered probabilities against the raw implied probability, for the single closing book with the highest measured margin in the archive, chosen by margin alone. The longshot outcome is annotated: this is where the transforms disagree most.

reversed. P3 was constructed on the assumption that a competently implemented Dixon–Coles model would show positive measured edge in at least some cells, against at least the softer books. It does not. The prediction was therefore not so much falsified as rendered untestable by this design, which is a limitation of the study rather than a finding about the transforms.

Table 5: The one cell in which the sign of measured ROI differs across transforms.

League	Bookmaker	Market	Matches	ROI (prop.)	ROI (Shin)	Spread
B1	Coral	1X2	214	0.0020	−0.0004	0.0184

7.4 Forecast accuracy against the benchmark

The model is beaten by the market comprehensively and almost uniformly. Table 6 gives the picture: mean model RPS of 0.2053 against benchmark RPS between 0.1987 and 0.1990, a gap of roughly 0.0065 in the market’s favour under every transform. The model achieves a lower RPS than the de-margined benchmark in *no* cell under power, Shin or the odds ratio, and in one cell of 91 under proportional normalisation. Around three-quarters of cells reject equal predictive accuracy at $q < 0.10$ after the Benjamini–Hochberg correction, in every case in the market’s favour.

Table 6: Model versus benchmark RPS by transform, with Diebold–Mariano statistics and Benjamini–Hochberg adjusted p -values.

Transform	Cells	Model RPS	Benchmark RPS	Diff.	Share fav. model	Share $q < 0.10$
Proportional	91	0.2053	0.1990	0.0063	0.0110	0.7582
Power	91	0.2053	0.1987	0.0065	0.0000	0.7473
Shin	91	0.2053	0.1988	0.0065	0.0000	0.7582
Odds-ratio	91	0.2053	0.1988	0.0065	0.0000	0.7582

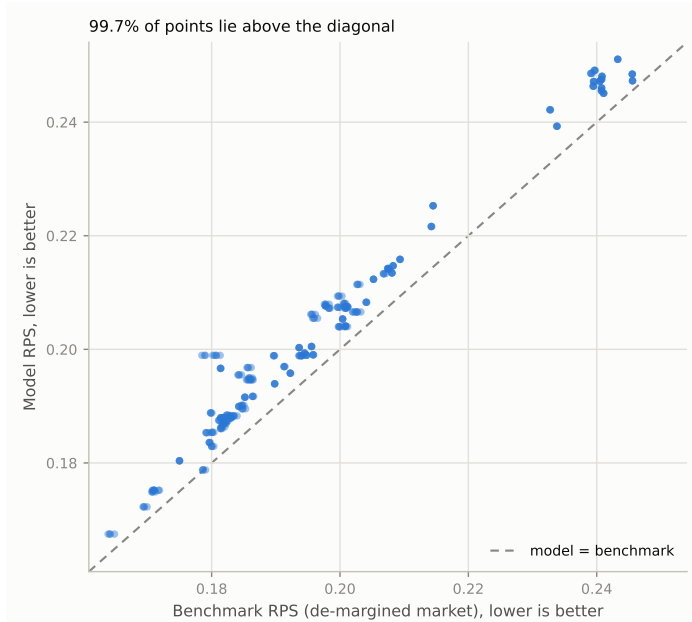


Figure 4: Per-cell model RPS against de-margined market RPS, one point per (cell, transform) pair. Almost every point lies above the diagonal: the model loses to the market in the same pattern Table 6 reports in numbers.

Economic returns agree. Mean ROI across cells ranges from -7.9% under proportional normalisation to -9.4% under the power method, and between 9% and 11% of cells show positive ROI at all. Exactly one cell of 91 has a bootstrap interval lying entirely above zero, which is fewer than the four or five that would be expected by chance across this many cells.

None of this is surprising and none of it indicates a defect. Dixon–Coles is a 1997 model with two parameters per team, no player information, no market data and no in-play adjustment, evaluated against closing prices that aggregate everything the betting market learned before kickoff. That it loses is the expected outcome. It is reported here because it determines what P3 can mean, and because a study that measured an edge without checking whether one existed would be reporting an artefact of its own arithmetic.

Per-cell ROI by transform (n = 91 cells each)

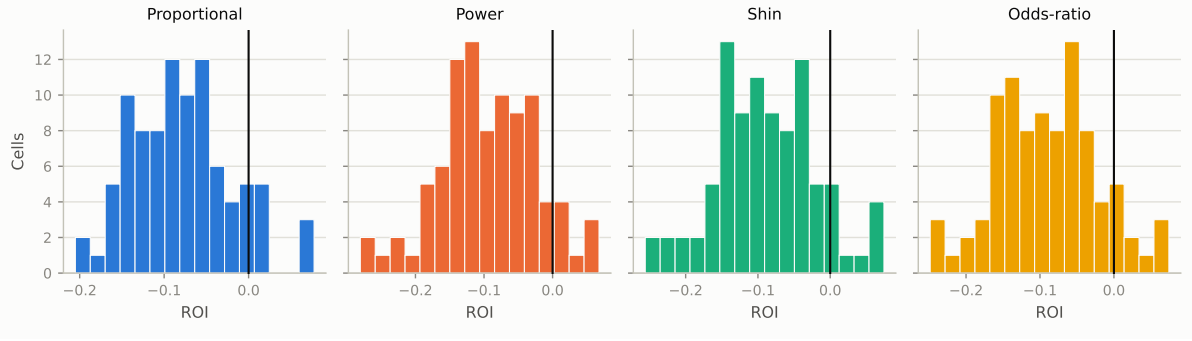


Figure 5: Distribution of per-cell ROI, one panel per transform, with a zero line on each. The mass sits on both sides of zero under every transform. The model has no measurable edge against any of the four benchmarks, which is the finding that governs how P3 should be read.

8 Discussion

The methodological claim survives and the headline claim does not.

That the four transforms diverge, and that they diverge in proportion to the margin being removed and more sharply where a longshot is present, is established here on 91 cells across 15 leagues and 14 bookmakers. A paper that de-margins a soft book by proportional normalisation and one that uses Shin’s method are not measuring the same benchmark, and the gap between them is largest exactly where a model’s claimed edge tends to concentrate. Reporting which transform was used, and ideally reporting a result under more than one, costs nothing and is not current practice.

What this study cannot show is that the choice flips a conclusion, because our model never reaches a conclusion worth flipping. The design held the model fixed so that any difference between arms would be attributable to the transform alone, which was the right control, but it assumed the fixed model would land near the market. It does not come close. Against that backdrop the single sign reversal we observe is noise around zero, and would be noise around zero whichever transform produced it.

The honest reading is that this work establishes the mechanism and bounds its size, while leaving the question it set out to answer open.

8.1 An exploratory extension that would settle P3

We do not rewrite the pre-registration, and we do not pretend a follow-up was registered. P3 asked whether transform disagreement can reverse the sign of a measured edge. The question remains open, because the model used here has no edge to reverse. What would close the gap is an explicitly post-hoc extension using either a stronger forecasting model (Rue and Salvesen’s dynamically updated ratings, or Karlis and Ntzoufras’s bivariate Poisson, Section 2), opening rather than closing lines, a deliberately softer bookmaker or segment, or a published model that already claims a positive edge. If that extension found a statistically credible edge that was significant under transform A and disappeared or reversed under transform B , the headline claim would be settled. That is a different study; the apparatus built here would run it unchanged.

9 Limitations

The model is fixed by choice. This study says nothing about whether a better model would beat the market. It asks only whether the measurement of any model’s edge depends on the transform. A model with a different error profile would concentrate its disagreements with the market at different prices, and the size, though not the existence, of the effect reported here is specific to this model.

Only two markets and only closing prices are used. Nothing here generalises to in-play markets, Asian handicaps, or opening lines.

Backtested returns are not achievable returns. There are no stake limits, no line movement between the moment a price is observed and the moment a bet could be struck, no commission, and no account restrictions. The ROI figures exist to be compared *across transforms*, which is what the design controls. Their absolute level is not a claim about what a bettor would have earned.

The decay parameter is weakly identified (Section 5.3). This does not threaten the comparison, which holds it fixed across arms, but it does mean the model should not be read as tuned.

The most consequential limitation is that the fixed model has no measurable edge anywhere in the sample. P3 was framed on the assumption that it would show positive edge in at least some cells against at least the softer books. It does not, in any cell, under any transform. Everything reported about P3 is therefore a statement about behaviour near zero rather than about whether the transform choice can overturn a real finding. P1 and P2 are unaffected, since both concern disagreement between benchmarks and neither requires the model to be any good. The exploratory extension in Section 8.1 is the natural next measurement; it is not part of this paper.

The calibration period is early. A fixed three-season burn-in and two-season calibration on a 33-season archive places the selection of ξ in 1996–97 for the longest-running leagues, up to 28 years before the evaluation period ends. The split was fixed in the design specification before any data was examined and was deliberately not revised once that consequence became visible.

10 Reproducibility

All code, the pre-registration, and the build for this paper are in the repository at <https://github.com/Bardakor/The-De-margining-Artifact>. The pre-registration cites the repository’s former name, which now redirects. That file is left byte-for-byte as committed, because its value rests on not having been edited after the fact. The study runs in three stages (**coverage**, then **calibrate**, then **evaluate**), in that order, with the pre-registration commit required between the second and the third. Every stochastic step takes an explicit seed and no global random state is touched. The tables in Section 7 are generated mechanically from the evaluation output rather than transcribed.

APPENDIX

Appendix A The five-term Murphy decomposition

Murphy’s vector partition [20] is usually quoted in three terms, $BS = REL - RES + UNC$, which is exact only when every bin holds a single distinct forecast value. Binning continuous forecasts leaves a residual. Expanding $\sum_i (p_i - o_i)^2$ within a bin around the bin means $\bar{p}, \bar{o}$ gives

$$(p_i - o_i) = (p_i - \bar{p}) + (\bar{p} - \bar{o}) + (\bar{o} - o_i), \quad (18)$$

and squaring leaves three squared terms plus one surviving cross term, $-2(p_i - \bar{p})(o_i - \bar{o})$. Summed, that is minus twice the within-bin *covariance* between forecast and outcome. It vanishes only when, inside every bin, higher forecasts carry no information about which events occurred.

The four-term form that stops at within-bin variance is therefore one term short. It is easy to miss because worked examples with one or two points per bin satisfy the zero-covariance condition by accident. On real continuous forecasts it does not vanish: measured on 4,000 simulated forecasts the covariance term accounted for 1.2% of the Brier score, which the four-term form silently misattributes to reliability and resolution. Including it drops the reconstruction residual from order 10^{-3} to machine zero. All five terms are reported, and the residual is computed explicitly so the identity can be checked rather than trusted.

Appendix B Implementation notes: from equations to code

The equations in Sections 3 and 5.1 describe what is computed. This appendix shows how, at the four points where the implementation choice is not a transcription of the mathematics but a decision in its own right. Each excerpt is quoted verbatim from the repository accompanying this paper (Section 10) and is shortened only by removing docstrings already quoted in the main text.

B.1 A shared root-finder for three different equations

Propositions 1–3 (Section 3.1) establish that the power, odds-ratio and Shin equations each have a unique root in a known bracket. Rather than three solvers with three tolerances, one bisection driver is shared, so that no transform’s estimated parameter is more or less converged than another’s purely as an artefact of solver choice:

```
def _bisect(objective: Callable[[float], float], lo: float, hi: float) -> float:
    low, high = lo, hi
    mid = 0.5 * (low + high)
    for _ in range(MAX_ITERATIONS):
        mid = 0.5 * (low + high)
        value = objective(mid)
        if abs(value) < TOLERANCE:
            return mid
        if value > 0.0:
            low = mid
        else:
            high = mid
    return mid
```

(*src/footy/market/demargin.py*)

`objective` is $f(k) - 1$, $g(c) - 1$, or $h(z) - 1$ from Section 3.1, each decreasing on its bracket by the propositions proved there, so the sign of `value` always identifies which half of the bracket contains the root. `TOLERANCE` is 10^{-12} and `MAX_ITERATIONS` is 200, both module constants shared by every call site rather than parameters a caller could vary per transform. Note what the function does *not* do: it never raises when a root is absent from the bracket, it simply returns whatever `mid` it last tried. That is safe here only because Section 3.1 proves a root always exists

in the brackets actually passed in. `power` and `odds_ratio` grow their upper bound geometrically until the objective’s sign confirms it brackets the root before calling `_bisect` at all, and `shin` passes the fixed bracket $[0, 1 - 10^{-12}]$ justified by Proposition 3’s boundary values.

B.2 Vectorising the Dixon–Coles gradient

Equations (13) sum s_m^λ and s_m^μ over the matches each team played at home or away. Written as a Python loop over matches this is correct but, over several thousand refits, slow enough to matter. Written as a scatter-add keyed by team index it is one call per sum and no per-match Python overhead:

```
grad_attack = np.bincount(data.home_idx, s_lam, minlength=n) + np.bincount(
    data.away_idx, s_mu, minlength=n
)
grad_defence = np.bincount(data.away_idx, s_lam, minlength=n) + np.bincount(
    data.home_idx, s_mu, minlength=n
)
grad_attack_free = grad_attack[:-1] - grad_attack[-1]
```

(*src/footy/fit/likelihood.py*)

`np.bincount(idx, weights, minlength=n)` returns, for each team index $0, \dots, n - 1$, the sum of `weights` at the positions where `idx` equals that index: exactly $\sum_{m: h(m)=i} s_m^\lambda$ when `idx = home_idx` and `weights = s_lam`. The last line is the chain-rule correction for the dropped attack parameter described after Equation (13): every team’s raw gradient is computed first, and only then is the last team’s gradient subtracted from the rest, because the subtraction is a property of the *parameterisation*, not of any individual team, and computing it after the scatter-add keeps that fact visible in the code rather than folded into per-team bookkeeping.

B.3 The block bootstrap, vectorised over resamples

Section 5.5 uses a stationary block bootstrap [23] for ROI confidence intervals. Its defining mechanism (continue the current block with probability $1 - 1/L$, or restart at a uniformly random position with probability $1/L$, wrapping past the end of the series) is inherently sequential in the resample *index*, but not across the 10,000 resamples drawn at once, so the sequential part is a Python loop over the (short) series length with every resample advanced together as a vector:

```
index = np.empty((n_resamples, n), dtype=np.int64)
index[:, 0] = starts[:, 0]
for t in range(1, n):
    advanced = (index[:, t - 1] + 1) % n
    index[:, t] = np.where(restart[:, t], starts[:, t], advanced)
```

(*src/footy/eval/inference.py*)

`restart` is a $(10,000, n)$ boolean array drawn once, true with probability $1/L$ per entry, and `starts` a same-shaped array of uniform random positions. The modulo in `advanced` is what makes a block that runs past the last observation wrap around to the first, which is the “stationary” in stationary bootstrap and what a naive block bootstrap without wraparound would get wrong at the series boundary. The loop runs n times, not $n \times 10,000$: every resample’s index at step t is computed in one vectorised statement.

B.4 Where the leakage check actually lives

Section 5.2 states that a matchday’s fit uses only matches that kicked off *strictly* before the cutoff, and that this is enforced in one place rather than trusted at every call site:

```
deltas = (pd.Timestamp(as_of) - stamps).dt.total_seconds().to_numpy(dtype=np.float64)
if np.any(deltas <= 0.0):
    n_bad = int((deltas <= 0.0).sum())
    raise ValueError(
        f"{n_bad} match(es) at or after the fit cutoff {as_of}; "
        "a fit may only use strictly earlier matches"
    )
```

(*src/footy/fit/decay.py*, inside the function that also computes each match’s age for the decay weight $w_m = \exp(-\xi t_m)$ of Equation (10))

Computing the decay weight and checking for leakage happen in the same function rather than two separate ones because every caller that needs one needs the other: any code path that asks “how old is this match” is, by construction, also asked to answer “could this match have leaked,” and putting both questions in one function makes it impossible to compute a match’s age without also having its non-negativity checked.

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